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Demand-and-Supply Imbalance Risk and Long-Term Swap Spreads

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1 Big picture

- **Question.** Why did long-term U.S. swap spreads become negative after the Global Financial Crisis, and how can we separate end-user demand shocks from intermediary supply shocks?
- **Core puzzle.** A swap spread is the fixed swap rate minus the like-maturity Treasury yield. After 2008, 30-year swap spreads became negative, even though a negative spread looks like a violation of the law of one price.
- **Mechanism.** Long-term swap spreads are determined by end users' net demand for swaps and by constrained intermediaries' supply of balance sheet.
- **Two compensation margins.** Intermediaries require compensation for using scarce capital and for bearing convergence risk: the risk that spreads move further against them because future demand/supply imbalances worsen.
- **Empirical proxy.** The key quantity proxy is *PD-UST-Net*, primary dealers' net position in coupon-bearing Treasuries.
- **Main empirical conclusion.** Demand and supply shocks both contribute meaningfully to swap spread volatility. Demand is especially powerful in forecasting short-horizon returns to swap arbitrage, while supply shocks are more persistent in the VAR decomposition.

2 Data, background, and measurement

2.1 Interest rate swaps and swap spreads

An interest rate swap exchanges fixed interest payments for floating payments linked to a short-term reference rate such as LIBOR, OIS, or SOFR. The fixed swap rate is set so the swap has zero value at inception. A swap spread is the fixed swap rate minus the like-maturity Treasury yield.

Detailed interpretation. Swap spreads are near-arbitrage objects. Their sign and term structure reveal the interaction between end-user hedging demand and intermediary constraints.

2.2 Market participants

End users include insurers, pension funds, commercial banks, non-financial corporations, mortgage hedgers, mortgage servicers, fixed-income asset managers, and relative-value investors. Intermediaries include broker-dealer swap desks and fixed-income hedge funds that absorb net end-user demand and hedge with Treasuries.

Detailed interpretation. Swaps are in zero net supply, so any net end-user desire to receive fixed must be matched by intermediaries willing to pay fixed.

2.3 Main variables

- *PD-UST-Net*: primary dealers' net long position in coupon-bearing Treasury securities.
- *PD-UST-Credit*: Treasury securities in through financing arrangements, used as a proxy for levered swap-spread trades by hedge funds and other intermediaries.
- *ST-OIS-Spread*: a proxy for balance-sheet costs or intermediary-capital shadow values.
- MBS duration, pension underfunding, corporate issuance, broker-dealer leverage, hedge-fund performance, term premium, implied volatility, CIP basis, and curve-noise measures are used to interpret demand and supply factors.

Detailed interpretation. The empirical design relies on both prices and quantities. Swap spreads are prices; *PD-UST-Net* is the observable quantity proxy.

3 Models and empirical specifications

3.1 Swap spread trade return

$$r_{t+1}^s = (s_t - m_t) - \frac{\delta}{1 - \delta}(s_{t+1} - s_t), \quad m_t = i_t^S - i_t^T.$$

Detailed interpretation. The current spread is the carry component; spread changes are the mark-to-market component. A trade may have attractive current carry but still be risky if spreads move further against the intermediary.

3.2 Intermediary optimization and constraints

$$\max_{x_t, o_t} E_t[w_{t+1}] - \frac{\alpha}{2} Var_t[w_{t+1}],$$

subject to

$$w_{t+1} = w_t + x_t r_{t+1}^s + o_t r_{t+1}^o, \quad \kappa_x |x_t| + \kappa_o |o_t| \leq w_t.$$

Detailed interpretation. Intermediaries are specialized, risk averse, and balance-sheet constrained. Both receive-fixed and pay-fixed positions consume scarce capital.

3.3 End-user demand and affine equilibrium

$$d_t = z_t^d + \gamma s_t, \\ s_t = A_0 + A_m z_t^m + A_d z_t^d + A_w z_t^w, \quad A_m > 0, \quad A_d < 0, \quad A_w > 0.$$

Detailed interpretation. Higher end-user demand to receive fixed pushes spreads down. Higher intermediary wealth raises spreads because intermediaries supply swaps more cheaply and need less compensation.

3.4 Expected returns to swap spread trades

$$E_t[r_{t+1}^s] = -\kappa_x \psi_t - \alpha V d_t.$$

Detailed interpretation. Expected returns contain a capital-cost component and a convergence-risk component. The second term is proportional to the intermediary position.

3.5 Sign restrictions and SVAR

Positive demand shock: $d_t \uparrow, |s_t| \uparrow$, Positive supply shock: $d_t \uparrow, |s_t| \downarrow$.

$$y_t = c + \sum_{l=1}^L C_l y_{t-l} + \xi_t, \quad y_t = [s_t^{(30)}, \hat{d}_t]'$$

Detailed interpretation. The SVAR maps reduced-form innovations in 30-year spreads and the quantity proxy into structural demand and supply shocks using the model's sign restrictions.

4 Identification and empirical design

Prices alone cannot identify demand and supply. The quantity proxy *PD-UST-Net* adds information about the scale and sign of intermediated swap positions. A demand shock changes quantity and spread magnitude together; a supply shock changes quantity and spread magnitude in opposite directions. The proxy is noisy because dealers also intermediate Treasury issuance, Fed purchases, and other positions, but this measurement error should attenuate the results.

5 Section-by-section study guide

5.1 Introduction

The introduction frames negative long-term swap spreads as a near-arbitrage puzzle and motivates a model with end-user demand, constrained intermediary supply, and convergence risk.

5.2 Background on swaps

This section explains why end users use swaps, why intermediaries hedge with Treasuries, and why OIS/SOFR spreads are cleaner no-arbitrage objects than LIBOR spreads.

5.3 Theory

The theoretical predictions to remember are signs: demand to receive fixed lowers spreads; intermediary wealth raises spreads; quantity predicts returns; the spread curve predicts maturity excess returns.

5.4 Empirical results

The empirical section validates *PD-UST-Net*, runs predictive regressions, estimates a sign-restricted SVAR, and interprets the latent demand and supply factors.

6 Tables and figures

6.1 Figure 1: Swap spreads and primary dealer's net Treasury position

Read: Panel A shows 30-year LIBOR swap spreads and primary dealers' Treasury positions flipping signs around 2009. Panel B compares LIBOR, OIS, and SOFR spreads. Panel C shows the downward-sloping post-GFC term structure.

Detailed interpretation. Figure 1 motivates the entire empirical strategy. The inverse movement between swap spreads and *PD-UST-Net* makes the quantity proxy plausible.

Economic message. The negative swap spread puzzle is a quantity-and-price regime change.

6.2 Table 1: Forecasting returns to the 30-year swap spread trade from 2009m1 to 2020m6

Read: Table 1 regresses 3-month and 12-month returns to the 30-year swap spread trade on *PD-UST-Net*, *PD-UST-Credit*, *ST-OIS-Spread*, and controls. The predictive power is strongest in the cleaner 2009–2018

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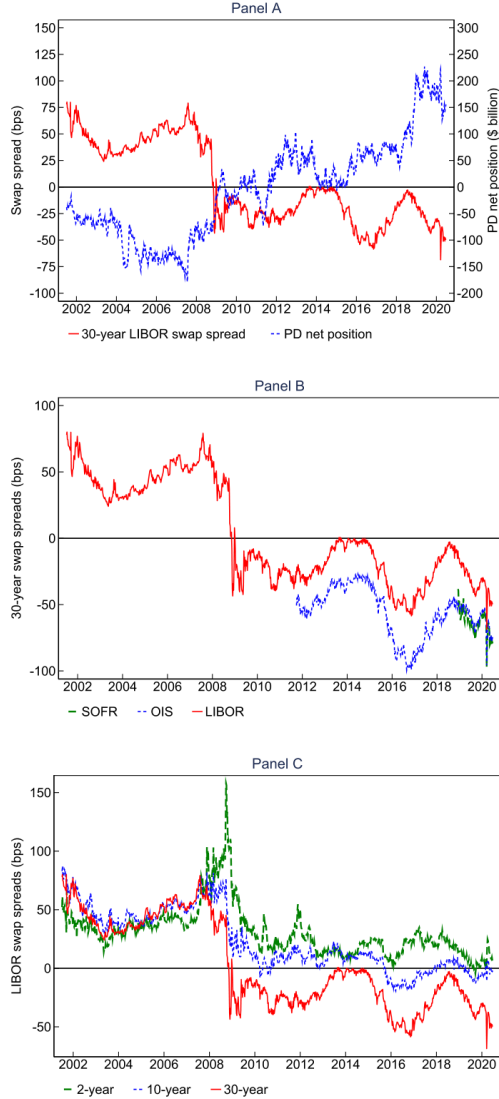


Fig. 1. Swap spreads and primary dealer's net Treasury position: Panel A shows the 30-year LIBOR swap spread and primary dealers' net position in coupon-bearing Treasury securities ($PD-UST-Net_t$). Panel B shows 30-year SOFR, 30-year OIS, and 30-year LIBOR swap spreads. Panel C shows 2-year LIBOR, 10-year LIBOR, and 30-year LIBOR swap spreads. Data are weekly and run from July 2001 to June 2020.

Table 1

Forecasting returns to the 30-year swap spread trade from 2009m1 to 2020m6: This table reports slope coefficients from regressions of the form

$$r_{t \rightarrow t+h}^{(30)} = \alpha + \beta_1 \cdot PD-UST-Net_t + \beta_2 \cdot PD-UST-Credit_t + \gamma_1 \cdot ST-OIS-Spread_t + \gamma_2' N_t + \epsilon_{t \rightarrow t+h}$$

We regress the returns to the 30-year swap spread trade on Primary Dealers' net position in nominal coupon-bearing Treasury securities ($PD-UST-Net_t$), Primary Dealers' outstanding balances of Treasury securities in through financing arrangements ($PD-UST-Credit_t$), the spread between the 3-month OIS rate and the 3-month T-bill rate ($ST-OIS-Spread_t$), and controls. Controls include the Federal Reserve's holdings of coupon-bearing Treasury securities, the 12-month moving average of the gross issuance of coupon-bearing Treasuries, and Primary Dealers' net position in corporate bonds. The holding period is equal to $h = 13$ weeks (3 months) in columns 1 to 4 and to $h = 52$ weeks (12 months) in columns 5 to 8. Data are weekly and run from January 2009 to June 2020 in Panel A and from January 2009 to June 2018 in Panel B. Newey and West (1987) t -statistics computed with 20 lags for 3-month return regressions and 78 lags for 12-month return regressions are reported in parentheses. We compute the associated p -values using the fixed- b asymptotic theory of Kiefer and Vogelsang (2005). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$h = 3\text{-month returns}$				$h = 12\text{-month returns}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Sample from 2009m1 to 2020m6								
$PD-UST-Net_t$	0.172 (0.47)	-1.064 (1.47)	0.243 (0.62)	-0.837 (1.12)	-1.747 (1.57)	-5.039*** (3.33)	-1.226 (1.04)	-4.053** (2.81)
$PD-UST-Credit_t$			-0.169 (1.09)	-0.392*** (2.78)			-1.243** (3.06)	-1.703*** (5.00)
$ST-OIS-Spread_t$	0.833 (0.22)	0.362 (0.09)	0.163 (0.04)	0.853 (0.23)	-17.155 (1.18)	-26.064** (2.45)	-22.096 (1.70)	-23.931** (2.61)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.001	0.043	0.010	0.086	0.075	0.141	0.231	0.327
N	599	599	599	599	599	599	599	599
Panel B: Sample from 2009m1 to 2018m6								
$PD-UST-Net_t$	-1.580*** (3.75)	-3.036*** (4.53)	-1.610*** (3.79)	-2.901*** (4.39)	-4.965** (2.72)	-7.437*** (4.19)	-5.142*** (3.45)	-6.752*** (5.72)
$PD-UST-Credit_t$			-0.200 (1.11)	-0.294** (2.41)			-1.179* (2.18)	-1.491*** (5.49)
$ST-OIS-Spread_t$	8.064** (2.20)	4.933 (1.57)	7.438** (2.14)	4.473 (1.47)	-1.440 (0.07)	-15.183 (1.14)	-5.133 (0.31)	-17.518 (1.60)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.132	0.178	0.149	0.205	0.216	0.379	0.348	0.526
N	495	495	495	495	495	495	495	495

sample.

Detailed interpretation. Controls matter because *PD-UST-Net* can also be affected by Treasury issuance and Fed holdings. Once these confounds are controlled for, the remaining component of *PD-UST-Net* behaves like the model’s quantity variable.

Economic message. Intermediated quantities forecast swap spread returns, consistent with convergence-risk compensation.

6.3 Table 2: Forecasting returns to 30-year and 10-year swap spread trades from 2009m1 to 2018m6

Read: Table 2 tests alternative dealer quantity proxies and compares 30-year and 10-year swap spread trades.

Detailed interpretation. Long-maturity Treasury measures work especially well because long-maturity Treasuries are natural hedges for long-maturity swap positions.

Economic message. Predictability is strongest where the model predicts the greatest convergence risk: long-dated swap spread trades.

Table 2
Forecasting returns to 30-year and 10-year swap spread trades from 2009m1 to 2018m6: This table reports the slope coefficients from regressions of the form

$$r_{t \rightarrow t+h}^{(n)} = \alpha + \beta_1 \cdot PD\text{-}UST\text{-}Net_t + \beta_2 \cdot PD\text{-}UST\text{-}Credit_t + \gamma_1 \cdot ST\text{-}OIS\text{-}Spread_t + \gamma_2' x_t + e_{t \rightarrow t+h}$$

for $n = 10$ and 30 . We regress returns to swap spread trades on alternative measures of Primary Dealers’ net position in Treasury securities (*PD-UST-Net*), Primary Dealers’ outstanding balances of Treasury securities in through financing arrangements (*PD-UST-Credit*), and the spread between the 3-month OIS rate and the 3-month T-bill rate (*ST-OIS-Spread*). We show results for our baseline net exposure measure (*PD-UST-Net*), a measure focusing on Treasuries maturing in 11 or more years (*PD-UST¹¹⁺-Net*), and a measure of levered exposure (*PD-UST-Lev*). Returns are for the $n = 30$ -year swap spread trade in Panel A and for the $n = 10$ -year swap spread trade in Panel B, and the holding period is equal to $h = 13$ weeks (3 months) in columns 1 to 4 and to $h = 52$ weeks (12 months) in columns 5 to 8. Data are weekly and run from January 2009 to June 2018. Newey and West (1987) t -statistics computed with 20 lags for 3-month return regressions and 78 lags for 12-month return regressions are reported in parentheses. We compute the associated p -values using the fixed- b asymptotic theory of Kiefer and Vogelsang (2005). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$h = 3\text{-month returns}$			$h = 12\text{-month returns}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Returns on the 30-year swap spread trade, $r_{t \rightarrow t+h}^{(30)}$						
<i>PD-UST-Net_t</i>	-1.610*** (3.79)			-5.142*** (3.45)		
<i>PD-UST¹¹⁺-Net_t</i>		-7.930*** (4.05)			-16.262* (2.24)	
<i>PD-UST-Lev_t</i>			-1.145*** (3.58)			-2.993** (2.53)
<i>PD-UST-Credit_t</i>	-0.200 (1.11)	-0.263 (1.50)	-0.298 (1.67)	-1.179* (2.18)	-1.283** (2.46)	-1.424* (2.36)
<i>ST-OIS-Spread_t</i>	7.438** (2.14)	8.482** (2.33)	7.050* (1.96)	-5.133 (0.31)	-5.058 (0.30)	-7.194 (0.43)
Adjusted R^2	0.149	0.164	0.133	0.348	0.228	0.252
N	495	495	495	495	495	495
Panel B: Returns on the 10-year swap spread trade, $r_{t \rightarrow t+h}^{(10)}$						
<i>PD-UST-Net_t</i>	-0.227 (1.38)			-1.017* (2.34)		
<i>PD-UST¹¹⁺-Net_t</i>		-1.890*** (3.71)			-2.523 (1.36)	
<i>PD-UST-Lev_t</i>			-0.170 (1.51)			-0.621 (1.68)
<i>PD-UST-Credit_t</i>	-0.134** (2.73)	-0.151*** (3.13)	-0.149*** (2.94)	-0.348** (3.25)	-0.361** (2.68)	-0.400** (3.15)
<i>ST-OIS-Spread_t</i>	1.385 (1.18)	1.810 (1.68)	1.340 (1.15)	-2.103 (0.65)	-2.337 (0.70)	-2.474 (0.74)

Table 3
ama-Bliss style forecasting regressions: This table reports the slope coefficients from regressions of the form

$$r_{t \rightarrow t+h}^{(n)} - r_{t \rightarrow t+h}^{(n')} = \alpha + \beta \cdot (s_t^{(n)} - s_t^{(n')}) + e_{t \rightarrow t+h}$$

where $n > n'$. In words, we regress the excess returns on a longer-dated swap spread trade over a shorter-dated trade ($r_{t \rightarrow t+h}^{(n)} - r_{t \rightarrow t+h}^{(n')}$) on the corresponding difference between longer- and shorter-dated swap spreads ($s_t^{(n)} - s_t^{(n')}$). The holding period is equal to $h = 13$ weeks (3 months) in columns 1 to 3 and to $h = 52$ weeks (12 months) in columns 4 to 6. Data are weekly and run from January 2009 to June 2020 in Panel A and from January 2009 to June 2018 in Panel B. Newey and West (1987) t -statistics computed with 20 lags for 3-month return regressions and 78 lags for 12-month return regressions are reported in parentheses. We compute the associated p -values using the fixed- b asymptotic theory of Kiefer and Vogelsang (2005). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$h = 3\text{-month returns}$			$h = 12\text{-month returns}$		
	(1)	(2)	(3)	(4)	(5)	(6)
	$r_{t \rightarrow t+h}^{(30)} - r_{t \rightarrow t+h}^{(1)}$	$r_{t \rightarrow t+h}^{(10)} - r_{t \rightarrow t+h}^{(1)}$	$r_{t \rightarrow t+h}^{(30)} - r_{t \rightarrow t+h}^{(10)}$	$r_{t \rightarrow t+h}^{(30)} - r_{t \rightarrow t+h}^{(1)}$	$r_{t \rightarrow t+h}^{(10)} - r_{t \rightarrow t+h}^{(1)}$	$r_{t \rightarrow t+h}^{(30)} - r_{t \rightarrow t+h}^{(10)}$
Panel A: Sample from 2009m1 to 2020m6						
$s_t^{(30)} - s_t^{(1)}$	2.550*** (3.03)			12.577*** (4.97)		
$s_t^{(10)} - s_t^{(1)}$		1.343*** (3.58)			2.988** (2.51)	
$s_t^{(30)} - s_t^{(10)}$			5.674*** (4.05)			21.333*** (6.58)
Adjusted R^2	0.077	0.101	0.140	0.422	0.163	0.454
N	599	599	599	599	599	599
Panel B: Sample from 2009m1 to 2018m6						
$s_t^{(30)} - s_t^{(1)}$	2.061** (2.33)			11.143*** (4.36)		
$s_t^{(10)} - s_t^{(1)}$		1.284*** (3.19)			2.940* (2.30)	
$s_t^{(30)} - s_t^{(10)}$			4.726*** (3.12)			18.623*** (5.62)
Adjusted R^2	0.063	0.098	0.120	0.381	0.168	0.384
N	495	495	495	495	495	495

6.4 Table 3: Fama-Bliss style forecasting regressions

Read: Table 3 shows that the spread-curve slope forecasts excess returns on longer-dated swap spread trades over shorter-dated trades.

Detailed interpretation. The slope of the swap spread curve is not just descriptive. It is a priced-risk object that reflects demand-and-supply imbalance risk by maturity.

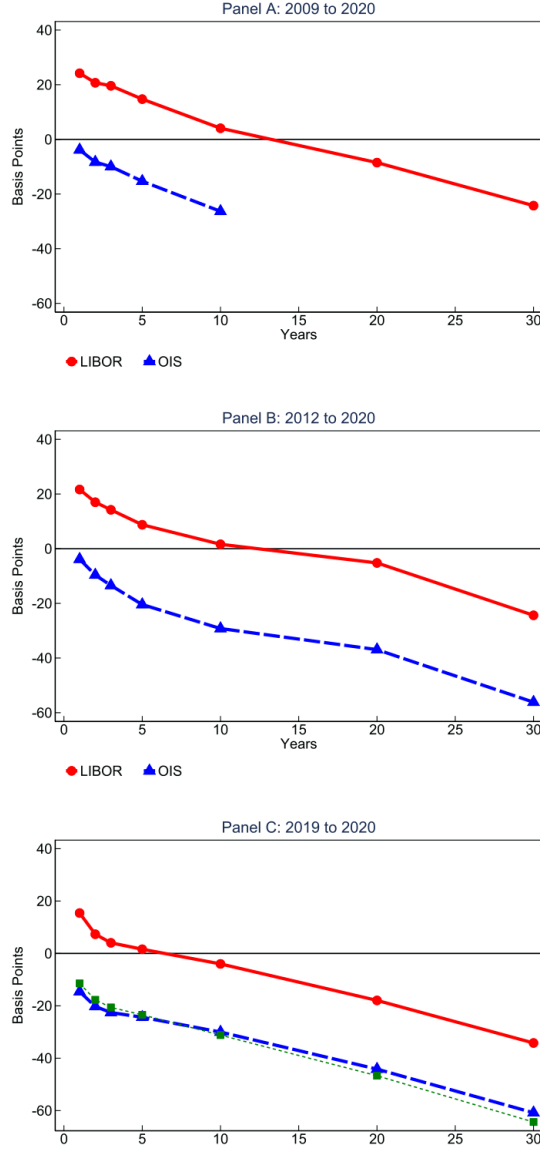
Economic message. The term structure of swap spreads contains expected-return information.

6.5 Figure 2 and Figure 3: Term structures and historical decomposition

Read: Figure 2 shows average term structures of LIBOR, OIS, and SOFR swap spreads. Figure 3 decomposes the 30-year swap spread into demand and supply contributions.

Detailed interpretation. Figure 2 shows that the negative-spread puzzle is a curve-level phenomenon. Figure 3 shows that the negative level after 2014–2018 cannot be explained by a single cause.

Economic message. Long-term swap spreads are a joint outcome of demand and intermediary supply.



(a) Term structures of swap spreads

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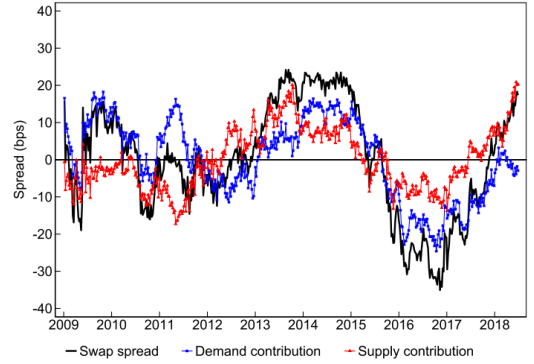


Fig. 3. Swap spread historical decomposition: This figure shows the 30-year swap spread minus its sample average (Swap spread), and the respective contributions of the demand factor (Demand contribution) and the supply factor (Supply contribution) in $s_t^{(30)} - E[s_t^{(30)}] = \hat{A}_d \hat{z}_t^d + \hat{A}_s \hat{z}_t^w$. The underlying data are weekly and run from January 2009 to June 2018.

(b) Historical decomposition

6.6 Table 4: Demand and supply drivers of 30-year swap spreads from 2009m1 to 2018m6

Read: Table 4 links the latent demand and supply components to observable economic proxies. MBS duration is strongly related to demand contributions; intermediary capacity proxies have more modest and noisier supply evidence.

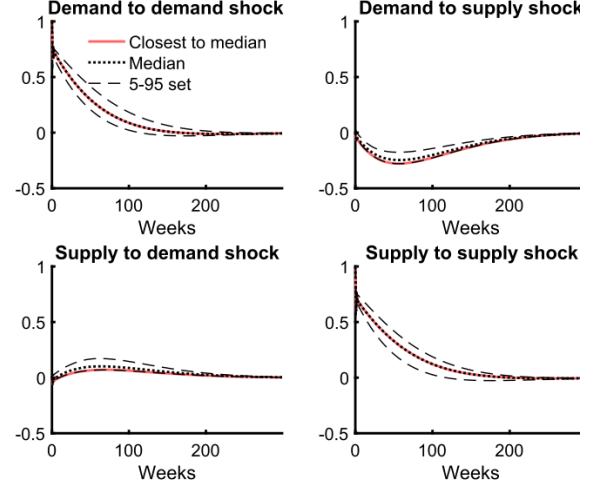
Detailed interpretation. Table 4 connects the latent VAR factors back to economically meaningful mechanisms, especially mortgage convexity hedging and pension duration management.

Economic message. Mortgage duration and pension underfunding help explain why end-user swap demand moves over time.

Table 4
Demand and supply drivers of 30-year swap spreads from 2009m1 to 2018m6: This table reports the slope coefficients from regressions of, respectively, the 30-year swap spread (columns 1 to 3), the demand contribution to spreads (columns 4 to 6), the supply contribution to spreads (columns 7 to 9), and PD-IST-Net, (columns 10 to 12) on the modified duration of the Barclays U.S. MBS index (*MBS-Duration*), the Klinger and Sundersen (2019) pension underlying factor (*Pension-UPR*), corporate bond issuance (*Corp-Issuance*), the broker-dealer leverage factor from Adrian et al. (2014) (*BD-Lev*), the logarithm of the HFRX Global Hedge Fund Index (*ln(HFR)*), the Adrian et al. (2011) term premium (*Term-From-ACM*), the VIXY Treasury volatility index (*UST-Implied-Vol*), the absolute deviation from 3-month covered interest-rate parity (computed as in Du et al. (2013)) averaged across the EUR-USD, GBP-USD, and JPY-USD currency pairs (*Ang-CIP-Basis*), and the Hu et al. (2013) yield curve fitting error (*UST-Curve-Noise*). Variables are in levels in Panel A and we control for a time trend (*t*) and in 13-week changes in Panel B. Data are weekly and run from January 2009 to June 2018. Newey and West (1997) *t*-statistics are reported in parentheses and are computed with 75 lags in Panel A and with 20 lags in Panel B. We compute the associated *p*-values using the fixed- β asymptotic theory of Kiefer and Vogelsang (2005). ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	$z_t^{(D)}$			$z_t^{(D)}$: Demand contrib.			$z_t^{(S)}$: Supply contrib.			PD-IST-Net		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Regressions in levels												
<i>MBS-Duration</i>	0.984*** (4.65)			6.297*** (7.49)			2.687 (1.66)			-11.523* (2.16)		
<i>Pension-UPR</i>	-1.448 (1.42)			-0.592 (1.16)			-0.857 (1.23)			-0.795 (0.38)		
<i>Corp-Issuance</i>	-0.007 (0.08)			-0.041 (0.49)			0.034 (0.62)			0.235 (0.65)		
<i>ln(BD-Lev)</i>		147.138*** (3.21)			43.275 (1.69)			103.863** (3.24)			186.131 (1.67)	
<i>ln(HFR)</i>		2.777*** (3.67)			1.894*** (4.41)			0.883* (2.22)			-3.232*** (3.08)	
<i>Term-From-ACM</i>			2.866 (0.70)			5.729* (2.18)			-2.863 (1.65)			-26.972* (2.29)
<i>UST-Implied-Vol</i>			-0.052 (0.05)			0.459 (0.53)			-0.491 (0.71)			-2.909 (0.84)
<i>Ang-CIP-Basis</i>			-0.631*** (3.53)			-0.330** (2.69)			-0.301*** (3.35)			0.107 (0.27)
<i>UST-Curve-Noise</i>			-0.232 (0.43)			-0.131 (0.39)			-0.102 (0.29)			0.097 (0.07)
Adjusted R^2	0.479	0.459	0.440	0.606	0.606	0.563	0.286	0.368	0.349	0.507	0.619	0.561
N	495	495	495	495	495	495	495	495	495	495	495	495
Panel B: Regressions in 3-month changes												
$\Delta_{13}MBS-Duration$				3.728*** (3.01)			0.666 (1.00)			-7.594** (2.44)		
$\Delta_{13}Pension-UPR$				-1.056 (1.46)			0.359 (1.00)			5.578*** (3.14)		
$\Delta_{13}Corp-Issuance$				-0.114** (2.36)			-0.025 (0.90)			0.200 (1.23)		
$\Delta_{13}ln(BD-Lev)$			-0.219 (0.01)			-13.906 (0.45)			13.687 (0.61)			86.305 (0.71)
$\Delta_{13}ln(HFR)$			1.097** (2.34)			1.364*** (4.39)			-0.267 (1.04)			-5.132*** (3.37)
$\Delta_{13}Term-From-ACM$				3.087 (1.06)			5.686*** (2.95)			-3.599 (1.41)		-26.014*** (3.47)
$\Delta_{13}UST-Implied-Vol$				1.030* (1.79)			0.229 (0.42)			0.801* (2.10)		1.766 (0.76)
$\Delta_{13}Ang-CIP-Basis$				-0.041 (0.46)			-0.037 (0.56)			-0.004 (0.09)		0.103 (0.45)
$\Delta_{13}UST-Curve-Noise$				0.007			0.247			-0.240		-1.523

Panel A: Impulse-responses for latent demand and supply factors



Panel B: Impulse-responses for observables

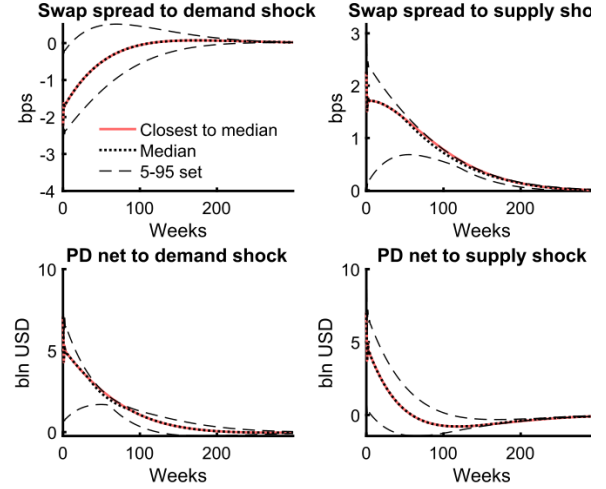


Fig. 4. Impulse response functions for latent demand and supply factors and for observables: These figures show the impulse response functions from our structural VAR. Median, “closest to median,” and 5th to 95th percent show the set of impulse responses that satisfy the identification restrictions. Data are weekly and run from January 2009 to June 2018. Panel A shows responses for the latent demand factor ($z_t^{(D)}$) and the latent supply factor ($z_t^{(S)}$). Panel B shows the responses for 30-year swap spreads ($s_t^{(30)}$) and quantile

6.7 Figure 4: Impulse response functions for latent demand and supply factors and for observables

Read: Demand shocks lower the 30-year spread and raise *PD-UST-Net*. Supply shocks raise the spread and raise *PD-UST-Net*. Supply responses are more persistent.

Detailed interpretation. Figure 4 is the dynamic counterpart of the sign restrictions. It shows not only impact signs but also adjustment speeds.

Economic message. Demand and supply shocks have different dynamic signatures.

7 Short summary

- The paper explains negative long-term swap spreads using a model with end-user demand and constrained intermediary supply.
- Primary dealers' net Treasury position proxies for the sign and scale of intermediated swap spread trades.
- *PD-UST-Net* predicts future returns on long-dated swap spread trades.
- A sign-restricted SVAR decomposes swap spreads into demand and supply components.
- Demand and supply shocks both explain spread variation; supply is more persistent, while demand is stronger for short-horizon return predictability.

8 Expanded conclusion

8.1 Core conclusion

Long-term swap spreads are shaped by the interaction of end-user hedging demand and constrained intermediary supply. Negative swap spreads are not simply a pricing mistake or one-dimensional balance-sheet-cost phenomenon.

8.2 Economic intuition

End users often want swaps for hedging reasons. Intermediaries must take the other side and hedge in Treasuries. This trade consumes capital and exposes intermediaries to convergence risk, so spreads adjust to compensate them.

8.3 Methodological contribution

The paper uses both price and quantity data. *PD-UST-Net* provides the quantity side that makes identification possible, and the sign-restricted SVAR maps observable movements into structural demand and supply factors.

8.4 Implications

Near-arbitrage spreads can remain open when intermediaries face capital constraints and convergence risk. This logic can apply to other long-dated near-arbitrage spreads such as CIP deviations and the CDS-bond basis.

8.5 Limitations

The quantity proxy is noisy, the structural VAR is set-identified, and the model abstracts from many institutional details. These limitations caution against over-interpreting any single episode, but they do not undermine the core price-plus-quantity logic.

8.6 Takeaway

Long-term swap spreads embed a demand-and-supply imbalance risk premium. Prices, quantities, and expected returns all point to the same message: end-user demand and intermediary supply jointly determine the level, sign, term structure, and dynamics of swap spreads.